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1) Addressing Modes

Given

PC = 4200H

R_1 = 1500 H

R_2 = 2500 H

RIx = 0800 H

SP = 3000 H.

Memory :-

Address	Content
1500H	45H
1502H	3200H
2500 H	18H
2800 H	3400 H
3000H	3600H
3200H	55H
3400H	60H
3600 H	77H
4204 H	200H

a) EA and operand value :-

1) Immediate Addressing :-

In immediate addressing, the operand itself igiven in the

operand = 200H

No effective address is required

EA = N/A, operand = 200H

2, Direct Addressing :-

$EA = 200H$

operand :- $M[200H]$

$EA = 200H$, operand $= M[200H]$ (not provided)

3, Indirect Addressing :-

for indirect addressing

$EA = M[200H]$

$EA = M[200H]$ (not provided.

4) Register Addressing :-

Assuming the register specified is R_1 :-

$R_1 = 1500H$

$EA = N/A$, operand $= 1500H$

If the instruction specified R_1, the operand would be 2500H

5, Register Indiret/Addressing :-

using R_1 :-

$EA = R_1 = 1500H$

Memory :- $M[1500H] = 45H$

$\therefore$ $EA = 1500H$, operand $= 45H$

6, Indexed Addressing :-

$EA = \text{Address} + Ix$

$EA = 200H + 0800H$

$EA = 0A00H$

operand $= M[0A00H]$ (not provided)

7, Relative Addressing :-

$$EA = PC + operand$$
$$EA = 4200H + 0200H$$
$$EA = 4400H$$
operand = M[4400H] (not provided)

8, Stack Addressing :-

Stack addressing
$$EA = SP = 3000H$$
$$M[3000H] = 3600H$$

$\therefore$ EA = 3000 H, operand = 3600 H

b, Which addressing mode required maximum memory access

A, Indirect Addressing

- first memory access is required to obtain the effective address
- second memory access is required to obtain the actual operand.

$\therefore$ Indirect addressing required the maximum memory access

C, Increasing order of execution speed.

A, Generally

Immediate < Register < Register indirect < Direct < Indexed < Relative < Indirect

Fastest -> slowest :-

Immediate -> Register -> Register Indirect -> Direct -> Indexed -> Relative -> Indirect

Stack addressing depends on the processor architecture, so it is normally considered separately

2) Processor Performance

A) Given

· Number of instructions:-

$IC = 900 \text{ million} = 900 \times 10^6$

$CPI = 2.8$

$f_1 = 2.4 \text{ GHz} = 2.4 \times 10^9$

$f_2 = 3.6 \text{ GHz} = 3.6 \times 10^9$

∴ Formula $\text{CPU Time} = \frac{IC \times CPI}{\text{Clock Rate}}$

1) Original execution time:-

Total clock cycles:-

$900 \times 10^6 \times 2.8$

$= 2.52 \times 10^9 \text{ cycles}$

∴ $T_1 = \frac{2.52 \times 10^9}{2.4 \times 10^9}$

$T_1 = 1.05$ seconds

2) New execution time:-

CPI 2.8.

$T_2 = \frac{900 \times 10^6 \times 2.8}{3.6 \times 10^9}$

$T_2 = \frac{2.52}{3.6}$

$T_2 = 0.70$ seconds

3, Speedup

$$\text{Speedup} = \frac{\text{Old Execution Time}}{\text{New Execution Time}}$$

$$= \frac{1.05}{0.70}$$

Speed up = 1.5

∴ The processor is 1.5 times faster

$(1.5 - 1) \times 100 = 50\%$

50% improvement.

4, If increasing clock frequency increases CPI by 10%.

Original CPI :- 2.8

$CPI = 2.8 + (10\% \times 2.8)$

$= 2.8 + 0.28$

$= 3.08$

$$T = \frac{900 \times 10^6 \times 3.08}{3.6 \times 10^9}$$

T = 0.77 seconds

5, Is increasing clock frequency always beneficial?

No, Increasing clock frequency can improve performance, but if is causes CPI to increase significally, the performance improved can be reduced

Thus :-

Higher clock frequency does not always guarantee proportional performal improvement.

3, Instruction Mix and CPU performance :-

Instruction type	Instruction Count	CPI
Arithmetic	5×10^6	2
memory Access	3×10^6	5
Branch	1×10^6	4
floating point	2×10^6	8

Clock frequency :-

$$f = 2.5\ GHz = 2.5 \times 10^9\ Hz$$

1, Total number of clock cycles :-

Total cycles = Σ (instruction Count $\times$ CPI)

Arithmetic :- $5 \times 10^6 \times 2 = 10 \times 10^6$

Memory :- $3 \times 10^6 \times 5 = 15 \times 10^6$

Branch :- $1 \times 10^6 \times 4 = 4 \times 10^6$

Floating point = $2 \times 10^6 \times 8 = 16 \times 10^6$

$\therefore$ Total cycle $= 45 \times 10^6$

2, CPU execution time :-

$$T = \frac{\text{Total Cycles}}{\text{Clock frequency}}$$

$$T = \frac{45 \times 10^6}{2.5 \times 10^9}$$

$$T = 0.018\ s$$

3, Average CPI :-

Total instruction count :- $5+3+1+2 = 11 \times 10^6$

$$CPI_{avg} = \frac{\text{Total cycles}}{\text{Total instruction}}$$

$$= \frac{45 \times 10^6}{11 \times 10^6}$$

$$CPI_{avg} = 4.09$$

4, Memory access CPI reduced from 5 to 3

New memory cycles :- $3 \times 10^6 \times 3 = 9 \times 10^6$

other cycles remain :- $10 + 4 + 16 = 30 \times 10^6$

New total cycles :- $30 + 9 = 39 \times 10^6$

New execution time :- $T_{new} = \frac{39 \times 10^6}{2.5 \times 10^6}$

$$T_{new} = 0.01568$$

5, Percentage improvement in performance.

Original time $T_{old} = 18ms$

New time $T_{new} = 15.6 ms$

Performance improvement :- $\frac{T_{old} - T_{new}}{T_{old}} \times 100$

$$= \frac{18 - 15.6}{18} \times 100$$

$$= 13.33 \%$$

4) Comparison of processors:

Given that

Processor A

$f = 4 GHz$, $CPI = 2.5$, $IC = 500$ million

Processor B

$f = 3 GHz$, $CPI = 1.8$, $IC = 550$ million

Processor C

$f = 5 GHz$, $CPI = 3.1$, $IC = 430$ million

formula :- $\text{CPU Time} = \frac{IC \times CPI}{\text{Clock Rate}}$

1) CPU execution time of each processor

Processor A.

$$T_A = \frac{500 \times 10^6 \times 2.5}{4 \times 10^9}$$

$$= \frac{1250}{4000}$$

$$T_A = 0.3125s$$

Processor B

$$T_B = \frac{550 \times 10^6 \times 1.8}{3 \times 10^9}$$

$$= \frac{990}{3000}$$

$$T_B = 0.33s$$

Processor C

$$T_C = \frac{430 \times 10^6 \times 3.1}{5 \times 10^9}$$

$$= \frac{1333}{5000} \Rightarrow T_C = 0.2666s$$

2, Ranking based on execution time.

$0.2666 < 0.3125 < 0.33$

$\therefore \quad C > A > B$

3, Speedup of faster over slowest.

Fastest = Processor C

Slowest = Processor B

$$\text{Speedup} = \frac{T_B}{T_C}$$

$$= \frac{0.33}{0.2666}$$

$$\text{Speedup} = 1.24x.$$

4, Processor B reduces instruction count by 10%.

Original count = 550 million

Reduction = 10% × 550 = 55 million

$550 - 55 = 495$ million

$$T'_B = \frac{495 \times 10^6 \times 1.8}{3 \times 10^9}$$

$$= \frac{891}{3000}$$

$$T'_B = 0.297s$$

5, Best processor recommendation:-

Processor C is the best choice.

5, Memory and Program Execution Trace

Initial memory:-

Address	Data
2500H	15H
2501H	25H.

$A \leftarrow A+B$

1, Contents of A and B after each instruction

1, LDA 2500H

$A = M[2500H] = 15H$

$A = 15H$

2, MOV B, A

$B \leftarrow A$

$B = 15H$

$A = 15H$

3, LDA 250H

$A = M[2501H]$

$A = 25H$

$\therefore A = 25H, B = 15H$

4, ADD B

$A \leftarrow A+B$

$A = 25H + 15H$

$A = 40H$

$B = 15H.$

5. STA 2502H

M[2502H] ← A

M[2502H] = 40H

A and B :- A = 40H, B = 15H

6. HLT

Processor stops.

2. What value is stored at 2502H?

from STA 2502H:-

M[2502H] = 40H

3. Which flags are modified?

ADD B

∴ 25H + 15H = 40H

∴ S=0, Z=0, AC=0, P=1, Cy=0

4. Execution Trace:-

Step	Instruction	A	B	Memory 2502H
1	LDA 2500H	15H	–	–
2	MOV B,A	15H	15H	–
3	LDA 2501H	25H	15H	–
4	ADD B	40H	15H	–
5	STA 2502H	40H	15H	40H
6	HLT	40H	15H	40H